ALICE L WALTON SCHOOL OF MEDICINE

Mitral Regurgitation and Mitral Valve Prolapse

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Adapted from: Mitral Regurgitation

Listen to this Brick
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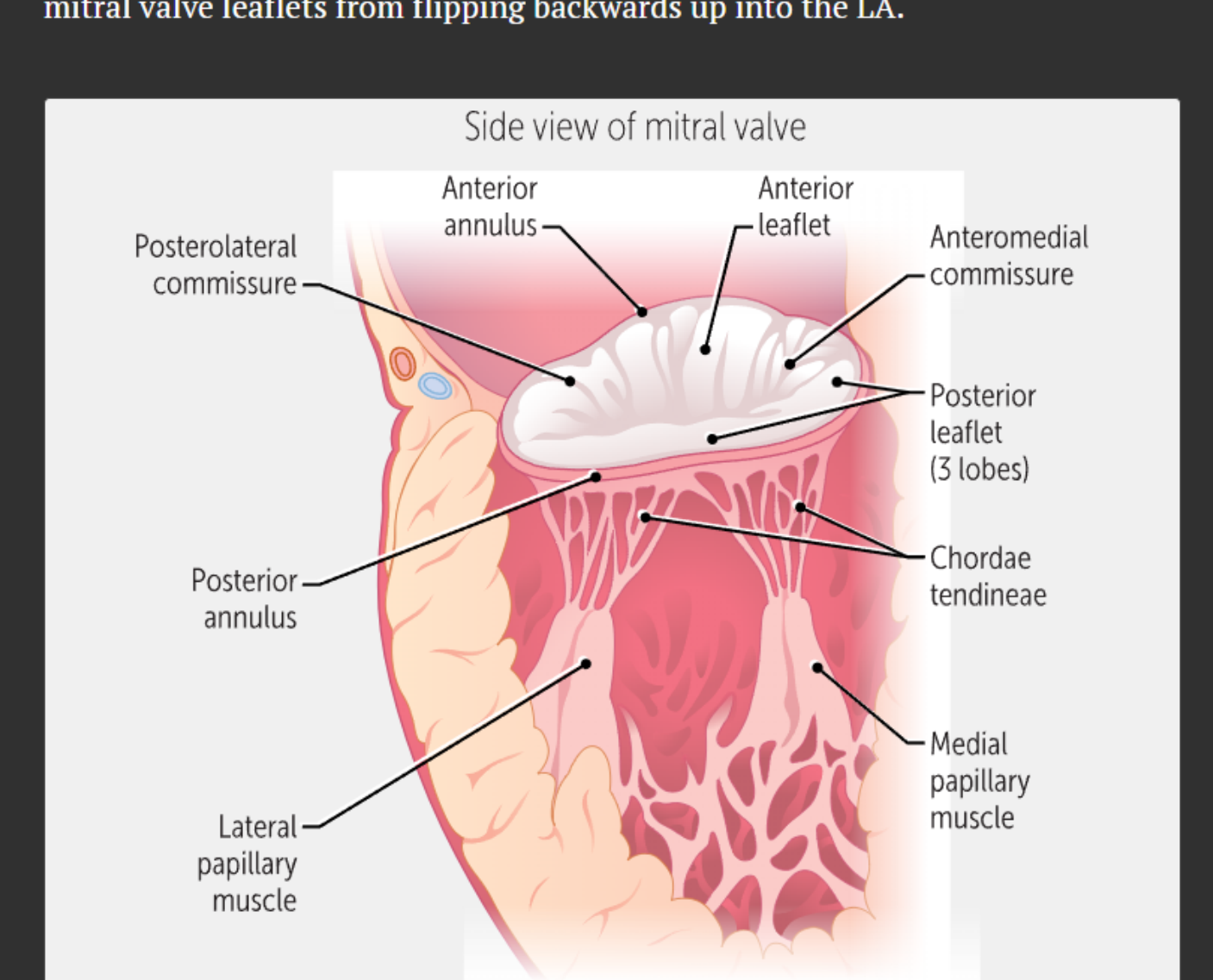
Learning Objectives (4) Versions

After completing this brick, you will be able to:

1. Describe the clinical presentation of Mitral Regurgitation and Mitral Valve Prolapse.
2. Explain the etiologies and pathophysiology of acute and chronic Mitral Regurgitation and Mitral Valve Prolapse.
3. Explain how Mitral Valve

myocardium and receive the same electrical stimulation. This pulls the chordae tendineae taut during the high pressure of ventricular systole and keeps the mitral valve leaflets from flipping backwards up into the LA.



QUIZ

Figure 1. Mitral Valve anatomy

MR is also more prevalent in those with advanced age.

When symptoms develop from MR, whether acutely or chronically, they are typically symptoms of heart failure: shortness of breath, lower extremity edema, and fatigue. When MR develops acutely (e.g. papillary muscle disruption), the left atrium and left ventricle do not have time to enlarge and adapt to the increased volume of blood. **Acute MR presents with severe pulmonary edema and possibly cardiogenic shock. Emergent surgical repair or replacement of the mitral valve is then required.**

In contrast, patients with chronic MR may be asymptomatic until late in the disease process because the LV has had time to adapt by eccentric hypertrophy. The presentation of chronic MR may be subtle, starting with generalized fatigue, dyspnea on exertion, and decreased exercise tolerance. Symptoms can progress to heart failure and dyspnea. They may also experience palpitations, as patients with chronic MR (or Mitral Stenosis) are more likely to develop atrial fibrillation due to atrial stretching and remodeling.

Why does severe, acute mitral regurgitation present with heart failure and cardiogenic shock?

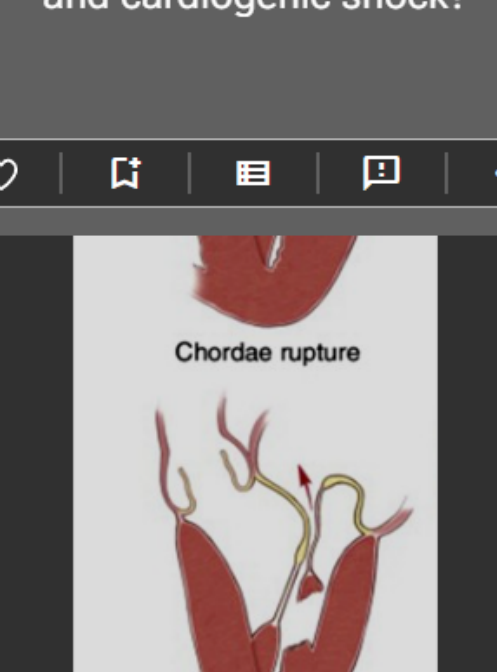
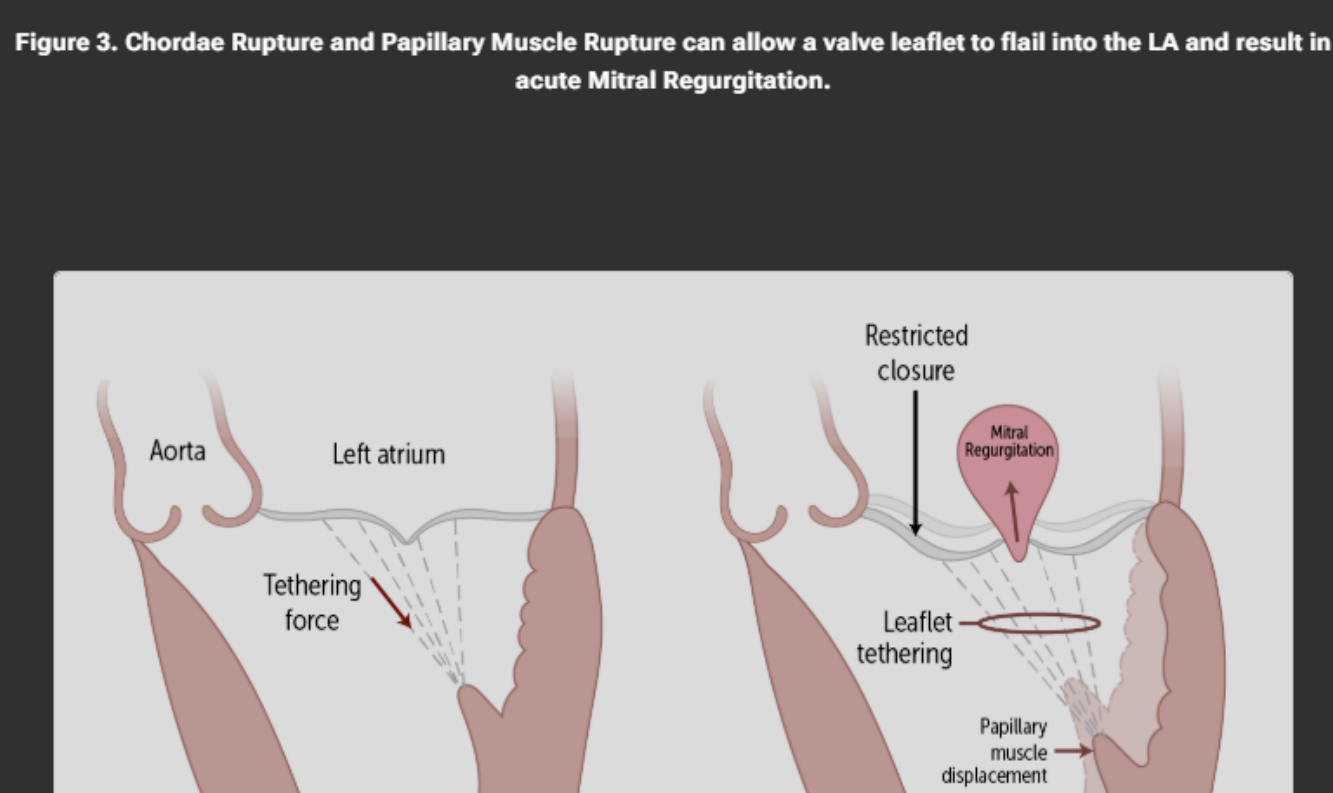


Figure 3. Chordae Rupture and Papillary Muscle Rupture can allow a valve leaflet to flail into the LA and result in acute Mitral Regurgitation.



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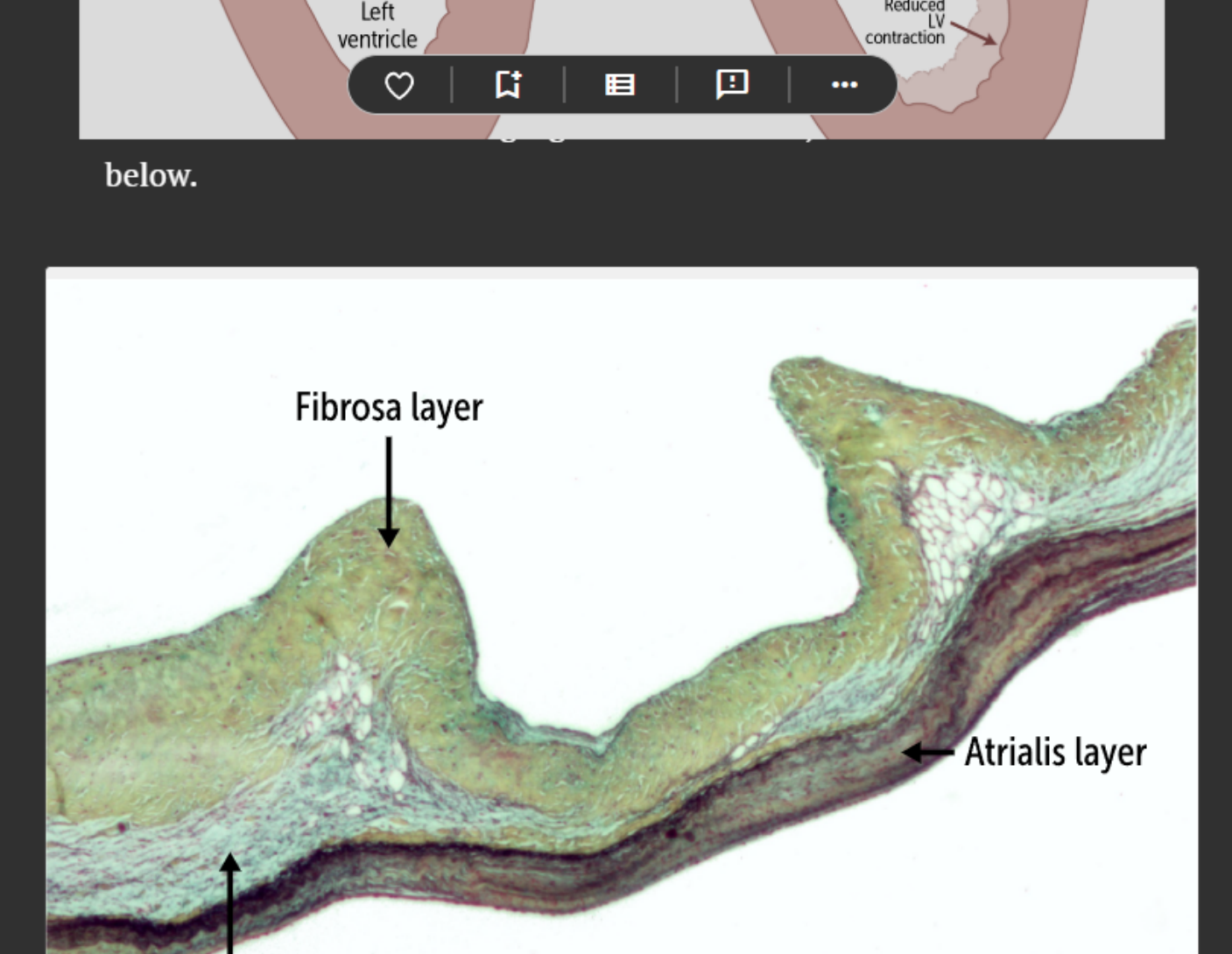


QUIZ

Figure 5. Myxomatous degeneration of the mitral valve

Additional causes of

through the insufficient mitral valve into the LA, called the **regurgitant volume.** Acute MR is very poorly tolerated due to the abrupt increase in LA volume and pressure. Blood flow from the pulmonary vessels into the LA is impeded, increasing pressure backwards into the pulmonary circulation, leading to pulmonary edema and dyspnea. The excess fluid may back up even further, overloading the right heart, leading to jugular venous distension (JVD) and peripheral edema.

In the setting of chronic MR, decreased blood ejection from the heart causes the LV to adapt by increasing the number of myocyte sarcomeres in series, lengthening the myofibrils and resulting in dilation. This **eccentric LV hypertrophy** is a form of pathologic ventricular dilation which tolerates higher volumes to help prevent pulmonary edema early in the course of disease. However, if LV dilation worsens and ventricular wall thickening cannot keep up, this cardiac remodeling becomes maladaptive and degrades LV function resulting in heart failure.

The LA will deliver the normal volume returning from the pulmonary veins plus the additional regurgitant volume into the LV. LV end-diastolic volume (LVEDV) will consequently increase **(Figure 10)**.

Increasing LA volume results in atrial dilation. This predisposes patients to atrial fibrillation in Mitral and Tricuspid Regurgitation due to volume overload and Mitral and Tricuspid Stenosis due to pressure overload.

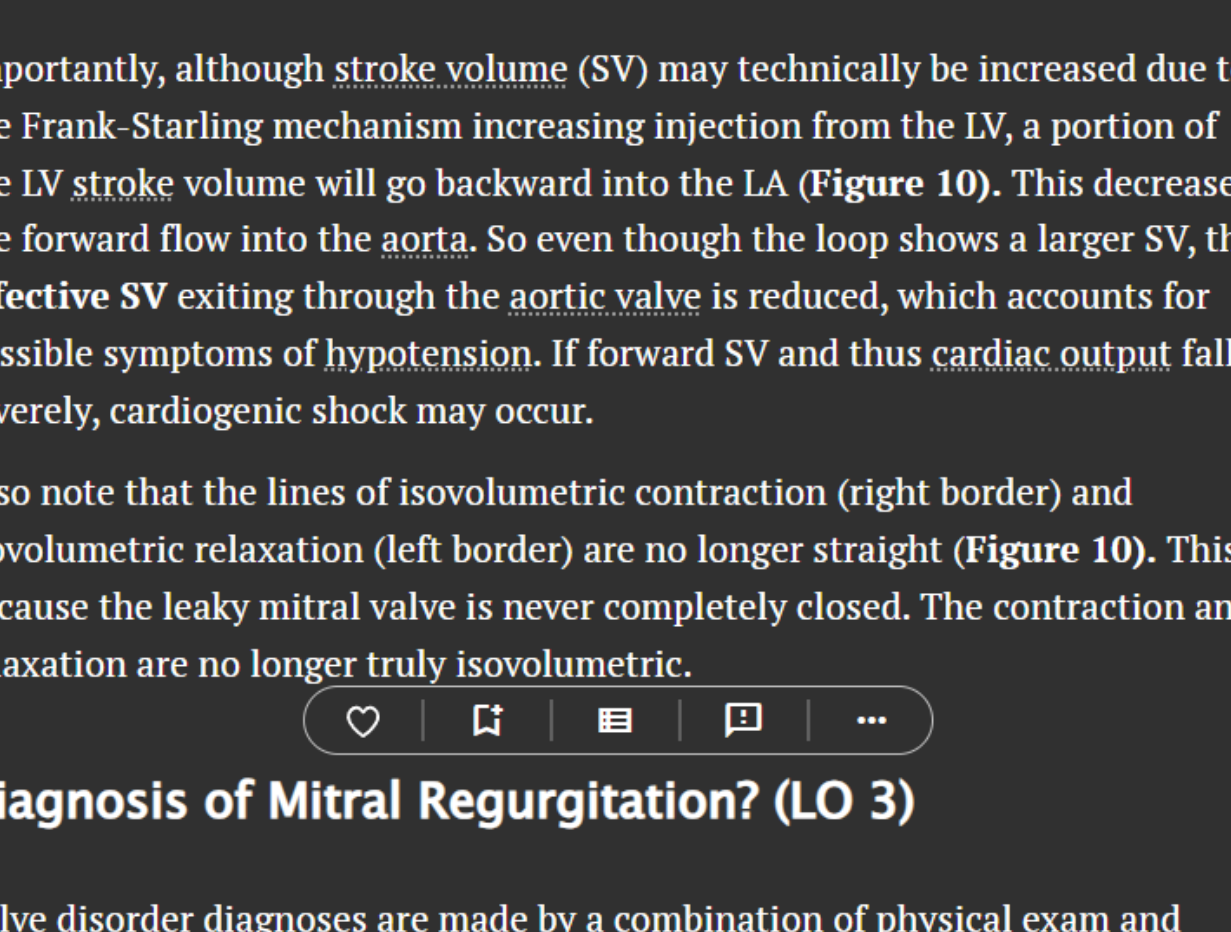


Figure 10. Pressure–volume loop of mitral regurgitation

Importantly, although stroke volume (SV) may technically be increased due to the Frank-Starling mechanism increasing injection from the LV, a portion of the LV stroke volume will go backward into the LA (**Figure 10).** This decreases the forward flow into the aorta. So even though the loop shows a larger SV, the **effective SV** exiting through the aortic valve is reduced, which accounts for possible symptoms of hypotension. If forward SV and thus cardiac output falls severely, cardiogenic shock may occur.

Also note that the lines of isovolumetric contraction (right border) and isovolumetric relaxation (left border) are no longer straight (**Figure 10).** This is because the leaky mitral valve is never completely closed. The contraction and relaxation are no longer truly isovolumetric.

Diagnosis of Mitral Regurgitation? (LO 3)

Valve disorder diagnoses are made by a combination of physical exam and echocardiography.

Physical Examination

Some patients (chronic) may present without symptoms, yet a heart murmur can be auscultated on routine exam. The murmur of MR is an **apical** systolic murmur. It is best heard over the apex of the heart: left fifth intercostal space at the midclavicular line. The murmur is usually heard best when the patient is positioned in the left lateral decubitus position. It typically **radiates posteriorly to the left axilla.** It tends to obscure S1 (the MV does not close properly), and sometimes S2, and lasts throughout systole so is classified as a **holosystolic** murmur **(Figure 11)**. However, when MR initially develops from MVP, it is heard as a mid systolic click with late systolic murmur **(Figure 12)**.

This murmur differs from the murmur of aortic stenosis (also systolic), which is a "whooshing" crescendo-decrescendo murmur, following the pattern of rising and falling LV pressure and velocity of blood flow across the narrowed valve during systole. It does not obscure S1, and it radiates to the carotids.

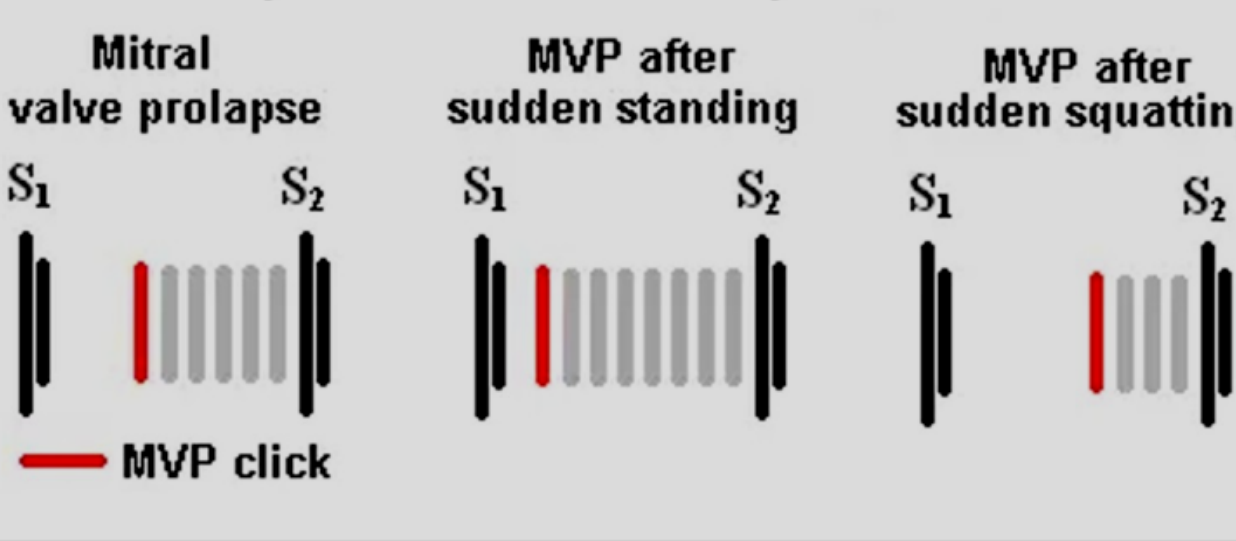


Figure 11. Typical murmur pattern in mitral regurgitation compared to aortic stenosis

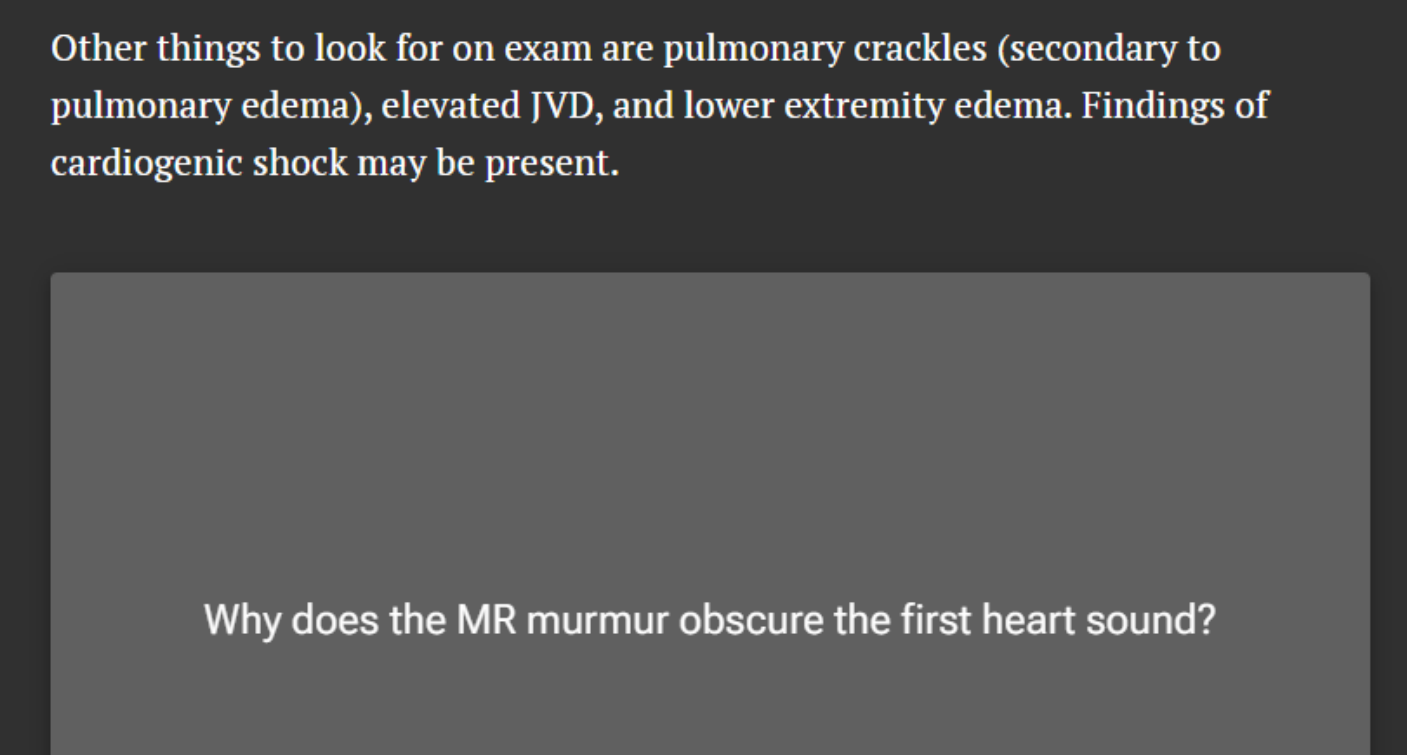

Mitral Valve Prolapse with Mitral Regurgitation:
Midsystolic click with late systolic murmur
Mitral valve prolapse
MVP after sudden standing
MVP after sudden squatting
S1
S2
MVP click

Figure 12. MR resulting from MVP results in a late systolic murmur, rather than holosystolic.

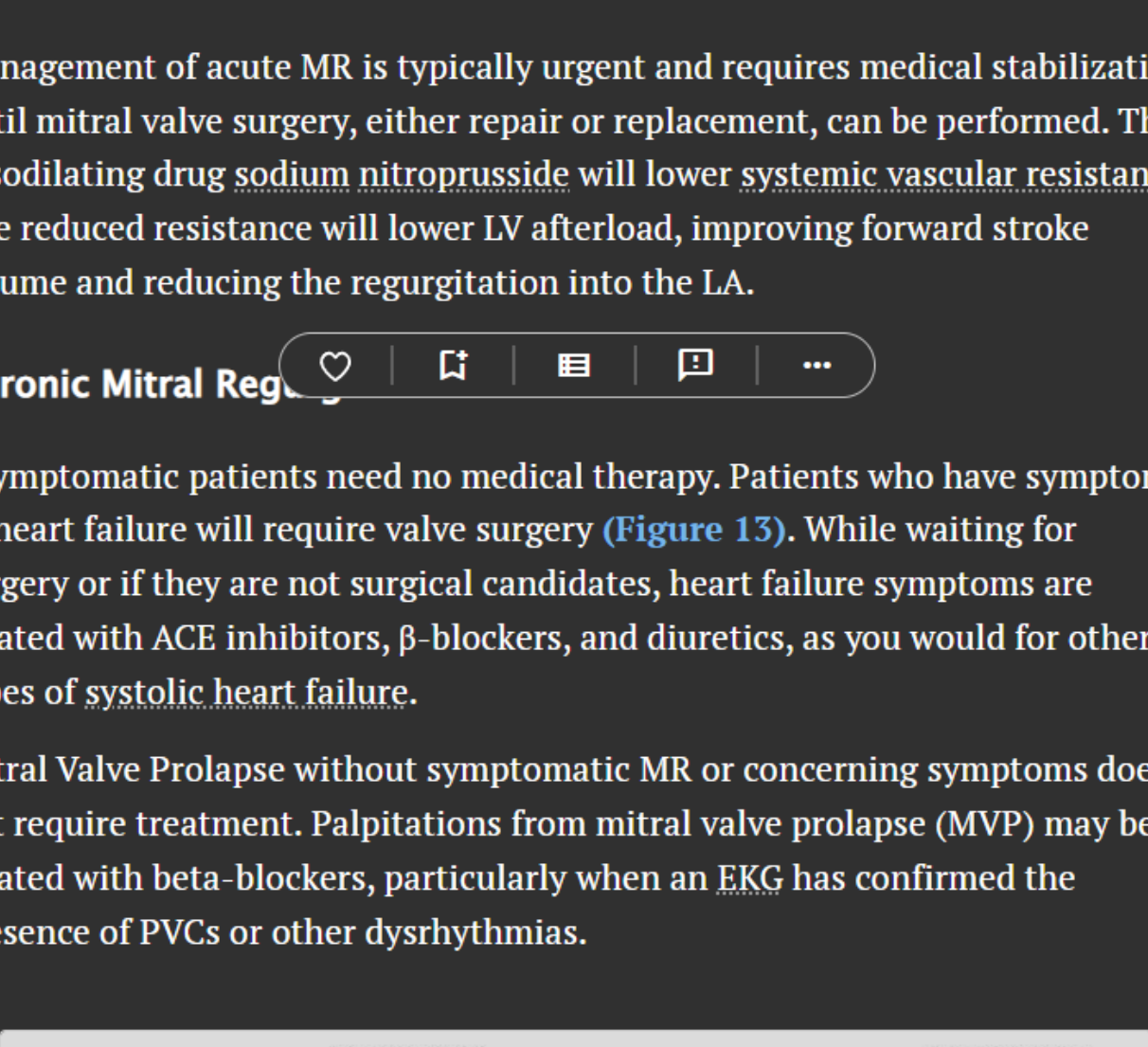
Other things to look for on exam are pulmonary crackles (secondary to pulmonary edema), elevated JVD, and lower extremity edema. Findings of cardiogenic shock may be present.

Why does the MR murmur obscure the first heart sound?

Imaging

The diagnosis of MR is confirmed by echocardiography. The severity of MR can be quantified by estimating the volume of regurgitant blood.

In the video below, note the Mitral Valve Prolapse followed by MR with a regurgitant jet crossing the mitral valve into the LA during ventricular contraction. Regurgitant blood exits the LV at the bottom left of the screen into the LA at the bottom right. In this parasternal long access view, the healthy aortic valve is seen a LV contraction.



Management of Mitral Regurgitation? (LO 4)

Management of MR varies depending on chronicity and symptoms.

Acute Mitral Regurgitation

Management of acute MR is typically urgent and requires medical stabilization until mitral valve surgery, either repair or replacement, can be performed. The vasodilating drug sodium nitroprusside will lower systemic vascular resistance. The reduced resistance will lower LV afterload, improving forward stroke volume and reducing the regurgitation into the LA.

Chronic Mitral Reg

Asymptomatic patients need no medical therapy. Patients who have symptoms of heart failure will require valve surgery **(Figure 13)**. While waiting for surgery or if they are not surgical candidates, heart failure symptoms are treated with ACE inhibitors, β-blockers, and diuretics, as you would for other types of systolic heart failure.

Mitral Valve Prolapse without symptomatic MR or concerning symptoms does not require treatment. Palpitations from mitral valve prolapse (MVP) may be treated with beta-blockers, particularly when an EKG has confirmed the presence of PVCs or other dysrhythmias.

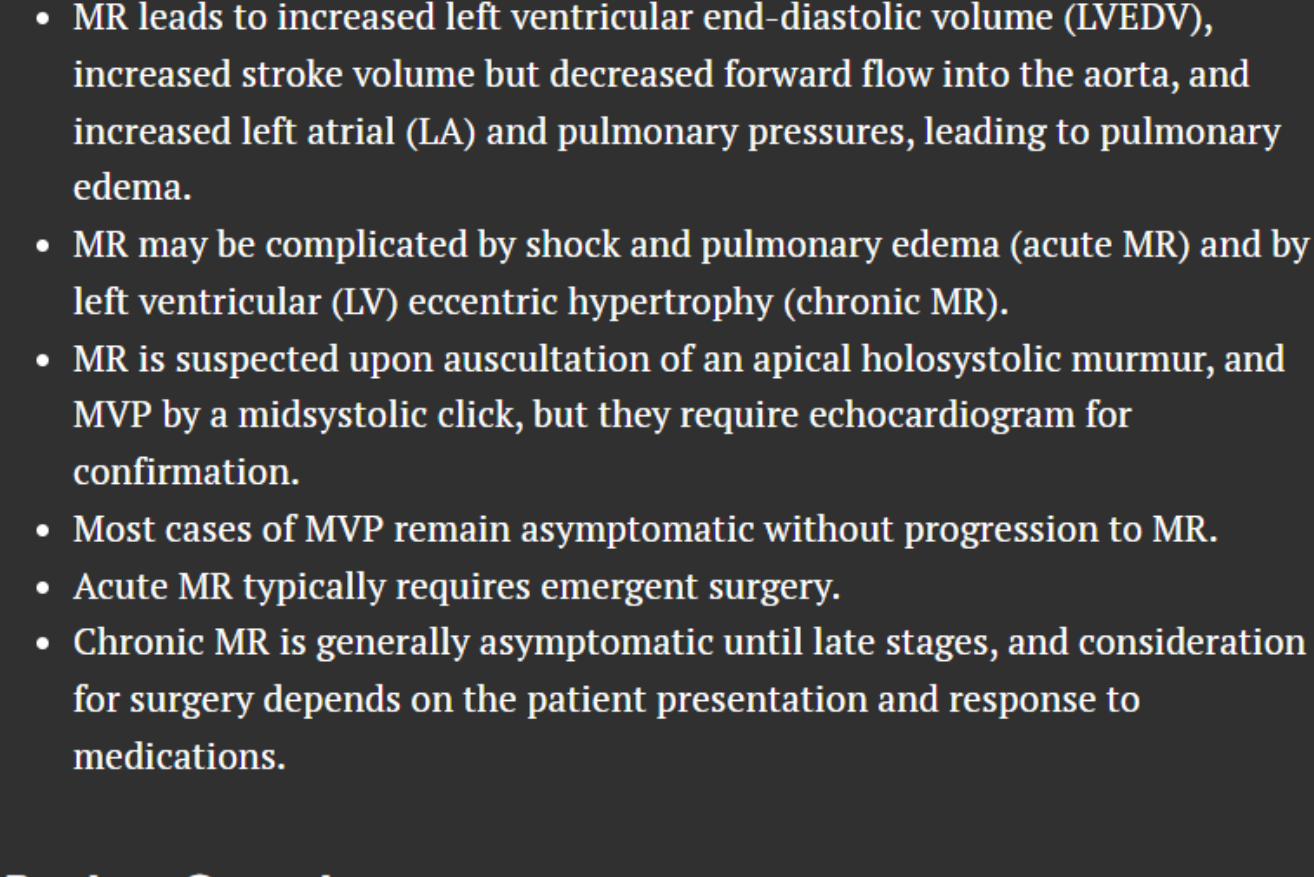


Figure 13. Mitral Va a mechanical valve.

ventricular wall.

- MR leads to increased left ventricular end-diastolic volume (LVEDV), increased stroke volume but decreased forward flow into the aorta, and increased left atrial (LA) and pulmonary pressures, leading to pulmonary edema.
- MR may be complicated by shock and pulmonary edema (acute MR) and by left ventricular (LV) eccentric hypertrophy (chronic MR).
- MR is suspected upon auscultation of an apical holosystolic murmur, and MVP by a midsystolic click, but they require echocardiogram for confirmation.
- Most cases of MVP remain asymptomatic without progression to MR.
- Acute MR typically requires emergent surgery.
- Chronic MR is generally asymptomatic until late stages, and consideration for surgery depends on the patient presentation and response to medications.

Review Questions

1. Which of the following is the least likely cause of acute mitral regurgitation?

A. ○ Acute myocardial infarction

B. ● Rheumatic heart disease

C. ○ Ruptured chordae tendineae

D. ○ Ruptured

surgery can be performed, although this therapy cannot definitively treat MR. β-Blockers (B) can help reduce symptoms of heart failure but are not a definitive treatment for MR. Balloon valvuloplasty (C) is a treatment for valvular stenosis (see other lessons).

3. How is the diagnosis of mitral regurgitation confirmed?

A. ○ Auscultation of the heart

B. ○ Nuclear Medicine Cardiac stress test

C. ○ Computed Tomography

D. ● Echocardiography

E. ○ X-ray

Hide Explanation

The correct answer is echocardiography (D), which confirms and determines the severity of MR. Auscultation of the heart (A) may reveal an apical systolic murmur characteristic of MR, but an echocardiogram is still needed for confirmation. NM stress test (B), Computed Tomography (C) and X-ray (E) do not evaluate valve function.

4. Patients with seve typically present

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